robstatpy: Python Wrappers for Modern Robust Statistical Estimation

Google Summer of Code 2026, Project Proposal

Organization: **R Project for Statistical Computing** · Submitted:

1. Project Title and Short Description

Title

robstatpy: Python Wrappers for the RobStatTM Robust Statistics R Package

Short Description

The RobStatTM R package (Maronna, Martin, Yohai & Salibian-Barrera, 2019) implements state-of-the-art robust estimators, including MM-regression, Rocke's S-estimator for covariance matrices, robust PCA via M-scale minimization, and Bianco–Yohai logistic regression, none of which have a direct equivalent in the Python ecosystem. This project will produce **robstatpy**, a Python library that makes the full range of RobStatTM functionality available to Python users through lightweight rpy2 wrappers, delivering a NumPy/pandas-native interface whose numerical outputs and graphical analytics reproduce the R originals. Every wrapper will ship with Python-style documentation modelled on the RobStatTM R man pages, a pytest validation suite, and reproducible Jupyter notebook examples.

2. Student Information

2.1. Contact Details

Field	Details
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Project Repo	github.com/Aakarsh751/robstattm-pyport-AG
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Communication	Zoom, Google Meet
Time Zone	Pacific Time (PT, UTC−7)

2.2. Academic Affiliation

Institution	Degree	Period
University of Washington, Seattle	B.S. in Computational Finance and Risk Management (CFRM)	Aug 2025 – Mar 2027
University of Michigan	B.S. in Data Science *(Dean's List; International Undergraduate Scholar)*	Aug 2023 – Aug 2025

I am currently a **junior** with an expected graduation of **March 2027**. I began my undergraduate studies in Data Science at the University of Michigan, where I was placed on the Dean's List and was awarded the International Undergraduate Scholarship. In August 2025, I transferred to the University of Washington to pursue a B.S. in Computational Finance and Risk Management, a programme that sits at the intersection of mathematical statistics, numerical computation, and financial theory, which is exactly the disciplinary blend required to bridge rigorous robust-statistics research and the Python data-science ecosystem.

2.3. Relevant Coursework

The following UW courses form the academic backbone of this project:

Course	Title and Relevance
Completed	
CFRM 425	*R Programming for Quantitative Finance*: hands-on R package development and financial modelling in R
CFRM 410	*Probability & Statistics for Computational Finance*: estimation theory, asymptotic methods, sampling distributions
CFRM 405	*Mathematical Methods for Quantitative Finance*: optimisation, convexity, and numerical root-finding
AMATH 482	*Computational Methods for Data Analysis*: dimensionality reduction, PCA, and spectral methods in Python
AMATH 352	*Applied Linear Algebra & Numerical Analysis*: matrix decompositions and numerical stability
Spring 2026 (in progress)	
CFRM 421	*Machine Learning with Application to Finance*: Python-based ML methods applied to financial data, directly mirroring the use-case of `robstatpy`
CFRM 586	*Financial Time Series Forecasting Methods*: modelling heavy-tailed financial returns, a primary application domain for robust estimation

2.4. Technical Skills

- **Programming languages:** Python, R, SQL, C++
- **Python ecosystem:** NumPy, pandas, scikit-learn, matplotlib, PyTorch, TensorFlow, Selenium; REST API development with FastAPI and Streamlit
- **R ecosystem:** `ggplot2`, `tidyr`, `dplyr`, `RobStatTM`, `robustbase`; DataCamp-certified in R Programming, Data Manipulation with dplyr, and Data Visualisation with ggplot2
- **Development tools:** Git/GitHub, Jupyter, RStudio, Docker, Anaconda
- **Finance:** Bloomberg Finance Fundamentals; quantitative equity and crypto-market analysis; portfolio risk modelling

2.5. Relevant Experience and Projects

IT Risk Management and Analyst Intern, Virginia Department of Motor Vehicles
Spearheaded a proof-of-concept initiative to evaluate AI and NLP for analysing call-centre interactions, assessing customer sentiment, call-resolution efficiency, agent performance, and fraud-related risk indicators. Built FastAPI back-ends and Streamlit front-ends to deploy and interact with large language models, enabling real-time inference and operational decision support. Designed and automated intelligent workflows using n8n to streamline repetitive tasks, and processed unstructured speech data to produce cost-benefit analyses guiding in-house versus outsourced AI deployment decisions. This role directly exercised the API-design, packaging, and data-pipeline skills that are central to `robstatpy`.

Undergraduate Research Volunteer, University of Michigan – Imitation Learning
Conducted research on Imitation Learning under faculty guidance, training reinforcement-learning and human-guided agents with OpenAI Gym. Implemented the DAGGER and HG-DAGGER algorithms from the primary literature to improve agent performance, adaptability, and learning efficiency, and evaluated all models rigorously using PyTorch. This experience demonstrates the ability to read formal algorithm specifications, translate them into well-tested, reproducible code, and verify results against published benchmarks, a skill directly applicable to wrapping the MM- and S-estimation routines in `RobStatTM`.

Deep Learning Algorithm for Asthenopia Detection Independently conceived and developed a deep-learning image-classification pipeline for the early detection of asthenopia (visual fatigue). The project was recognised with a **Gold Award at the Indian Science and Engineering Fair** and achieved a top-100 national selection, providing evidence of the capacity to drive an independent research project from initial conception through to a peer-recognised outcome.

2.6. Why My Background Aligns with This Project

My training in Computational Finance and Risk Management (CFRM) is uniquely suited to this project. Robust statistics is not an abstract topic for me: it is central to the quantitative methods I am now beginning to study and apply. In CFRM 410 (Probability and Statistics), I learned estimation theory and asymptotic analysis that will allow me to understand the underpinnings of robust MM-estimators and S-Estimators that are covered in Maronna et al. (2019). In CFRM 425 (R Programming), I developed R packages and gained hands-on familiarity with the object system that `RobStatTM` uses internally. In CFRM 405 (Mathematical Methods), I studied the optimisation numerical algorithms that drive iterative robust estimators. AMATH 482 (Computational Methods for Data Analysis) gave me direct experience implementing PCA and dimensionality reduction in Python, the exact techniques that `pcaRobS` robustifies.

This combination means I understand *both* the R code base and its statistical foundations, and the Python ecosystem that `robstatpy` targets. I am not just a software engineer wrapping an unfamiliar library; I have the theoretical knowledge to verify correctness, interpret results, and extend the project into downstream applications such as robust portfolio construction and risk analysis (RPCRA).

2.7. Relationship with Mentors

I have been in active communication with **Prof. R. Douglas Martin** since **January 2026**, attending weekly meetings to discuss the project scope, the Python API design, and the technical challenges of bridging R's S3 object system with Python's language model. These conversations have directly shaped the implementation strategy presented in this proposal. I look forward to further engaging with the full mentorship team, Prof. Matias Salibian-Barrera and Brian G.

Peterson, as the project progresses.

2.8. Mentors

Name	Role	Contact
Prof. R. Douglas Martin	Primary mentor	martinrd3d@gmail.com
Prof. Matias Salibian-Barrera	Co-mentor	matias@stat.ubc.ca
Brian G. Peterson	Co-mentor	brian@braverock.com

2.9. Time Commitment and Availability

I am applying for the **large project size (350 hours)**. I am fully dedicated and available for the entire duration of the GSoC coding period (late May through mid-November 2026), with **30–35 hours per week** committed to `robstatpy` development. I will continue attending weekly meetings with Prof. Martin and will be responsive to all three mentors via email, Zoom, and GitHub.

3. Benefits to the Community

The robust-statistics methods described in `RobStatTM` including MM-regression, Rocke's S-estimator, robust PCA via M-scale minimization, among others, are the algorithmic companions to *Robust Statistics: Theory and Methods* (Maronna, Martin, Yohai & Salibian-Barrera, 2nd ed., 2019), a leading modern textbook in the field. Today, a Python practitioner who wants to reproduce a textbook example or apply these methods in their research or applications faces two unappealing choices: install and learn R, or re-implement algorithms from scratch in Python.

`robstatpy` removes this barrier by providing a pip[1]-installable Python package that:

1. **Exposes the full `RobStatTM` API** through `rpy2` wrappers, accepting NumPy arrays and pandas DataFrames and returning native Python objects.
2. **Guarantees numerical fidelity:** every output agrees with the R original to machine precision, because the R code itself executes underneath.
3. **Lowers the maintenance burden:** because the wrappers delegate to R, any upstream improvement to `RobStatTM` is automatically available in Python without duplicating development effort.
4. **Reproduces figures and numerical summaries** using `matplotlib` and `plotnine`[2] so that Python users can produce visualisations that closely mirror the R originals.
5. **Provides a comparative study** of wrapper-based versus direct-porting strategies, producing benchmarks and guidelines that will inform future R-to-Python interoperability projects.

Who benefits:

- *Data scientists, ML engineers, and Python-only practitioners* who need a rich toolkit of modern robust estimation methods without learning R. `robstatpy` provides a comprehensive,

[1] `pip` is the standard Python package installer. Running `pip install robstatpy` downloads and installs the package and all its dependencies automatically.

[2] `plotnine` is a Python implementation of R's `ggplot2` grammar-of-graphics, enabling near-identical plot syntax in Python. See https://plotnine.org/ and https://realpython.com/ggplot-python/.

self-contained Python library: users call familiar Python functions with NumPy arrays and pandas DataFrames and never need to write, read, or understand any R code.

- *Students and researchers* who study the Maronna et al. textbook and want to apply the methods in Jupyter rather than RStudio.

4. Related Work

4.1. Existing Python Robust-Statistics Packages

- **scikit-learn** provides `MinCovDet` (fast MCD) and `HuberRegressor`, but lacks MM-regression, S-estimators for covariance, and robust PCA beyond standard PCA.
- **statsmodels** offers `RLM` (M-regression with Huber or Tukey bisquare weights) but does not implement the two-stage S+M refinement that defines MM-estimation, nor `lmrobdet.control`-style tuning.
- **RobPy** (Leyder et al., 2024; arXiv:2411.01954) is a comprehensive Python robust-statistics package offering MM-regression, S-regression, fast LTS, MCD-based covariance estimators (including cellwise MCD), ROBPCA, and cellwise outlier detection, all built as native re-implementations drawing from `robustbase`, `rrcov`, and `cellWise`. However, it does not expose the `RobStatTM` function set: the DCML regression family, robust logistic regression, additional multivariate scatter estimators, and the SM-PCA variant, along with the full `lmrobdetDCML`/`lmrobdetMM` tuning infrastructure codified in Maronna et al. (2019), remain without a Python equivalent.
- **robustats** (PyPI) provides C-backed univariate estimators (median, MAD, Qn) but nothing multivariate.

5. Proposed Work

`robstatpy` pursues a clear, phased strategy: first expose the complete `RobStatTM` function set to Python users via lightweight `rpy2` wrappers for immediate, numerically exact access; then provide alternative approaches for flexibility and portability. This section describes the primary wrapper approach, presents code examples for key functions already implemented, and reports validation results confirming numerical equivalence with R.

5.1. Why Wrappers Are the Right First Step

The `RobStatTM` package is **actively maintained** by its original authors, the same researchers who proved the theoretical properties of these estimators. A wrapper-based approach has three decisive advantages over full re-implementation:

1. **Exact numerical equivalence.** Because the R code executes underneath, Python users get bit-identical results. This is essential for textbook reproducibility and for cross-validating future pure-Python ports.
2. **Zero duplication of maintenance.** When the R authors fix a bug or add an estimator, the Python wrapper inherits the improvement immediately. A pure-Python re-implementation creates a second code base that must be independently maintained and validated.
3. **Speed of delivery.** Wrapping a function requires writing conversion glue, tests, and documentation, not re-deriving the algorithm. This makes it feasible to expose the *entire*

API within a single GSoC period, rather than a subset.

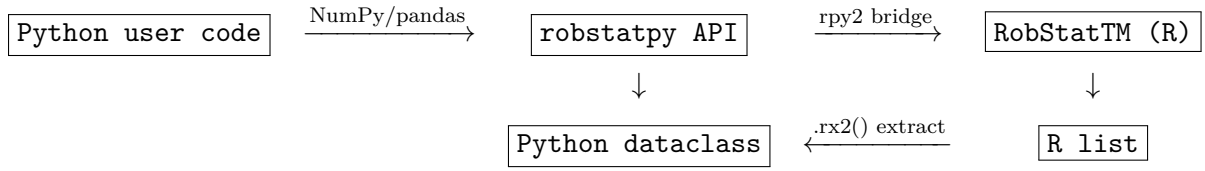
The project will *also* produce pure-Python re-implementations, beginning with `locScaleM` and `scaleM` and extending toward a complete native Python port of the `RobStatTM` library. The wrapper layer serves as the validated ground truth against which every pure-Python function is tested: once a ported function passes numerical equivalence checks against the rpy2 wrapper, it can replace the wrapper call, progressively removing the R dependency. This dual-track strategy, wrappers first for correctness and coverage, then native ports for independence and performance, provides the most robust path to a fully self-contained Python library while ensuring that no accuracy is sacrificed along the way.

5.2. Strategy A: rpy2 Wrapper Layer (Primary Approach)

Every public `RobStatTM` function will be wrapped in a Python function that:

1. Accepts NumPy arrays and/or pandas DataFrames.
2. Converts inputs to R objects via `rpy2.robjects` and `numpy2ri`/`pandas2ri` converters.
3. Calls the R function inside `RobStatTM`.
4. Extracts named elements from the returned R list using `.rx2()`.
5. Returns a Python dataclass (or named dict) whose fields match the R return values.

Architecture diagram:



Key design decisions:

- **Global conversion context.** `rpy2` 3.6+ stores conversion rules in a `contextvars.ContextVar`, which is lost across Jupyter's async task boundaries. We patch this by calling `set_conversion(default_conv` at import time, ensuring conversions persist across cells.
- **R dot-to-Python underscore mapping.** `importr()` automatically maps `lmrobdet.control` to `lmrobdet_control`. We document both names in every docstring.
- **Lazy R package loading.** `RobStatTM` is loaded via `importr` only on first use, so importing `robstatpy` in a pure-Python context (e.g. for type hints or documentation builds) does not require R.

The complete `locScaleM` and `scaleM` wrapper implementations, pure-Python re-implementations, and detailed numerical test results are provided in the companion notebook in the project repository: robstatpy_comparison_rpy2.ipynb.

Code Example: rpy2 Wrapper for lmrobdetMM

Listing 1: rpy2 wrapper for lmrobdetMM (regression; demonstrated with mopt family, efficiency=0.95)

```
def lmrobdet_mm(formula, data, family="mopt", efficiency=0.95):
    """Robust MM-regression via RobStatTM::lmrobdetMM().

    Parameters
    ----------
```

```
    formula : str
        R-style formula, e.g. 'zinc ~ copper'.
    data : pd.DataFrame
        Data frame (converted to R via pandas2ri).
    family : str
        Psi function family for IRLS refinement.
    efficiency : float
        Target asymptotic efficiency.

    Returns
    -------
    dict with 'coefficients', 'scale', 'residuals',
         'fitted_values', 'rweights', 'r_squared'.
    """
    from rpy2.robjects import pandas2ri
    pandas2ri.activate()
    r_df = ro.conversion.py2rpy(data)
    ctrl = robstattm.lmrobdet_control(
        family=family, efficiency=efficiency)
    fit = robstattm.lmrobdetMM(
        ro.Formula(formula), data=r_df, control=ctrl)
    return {
        "coefficients":  np.array(fit.rx2("coefficients")),
        "scale":         fit.rx2("scale")[0],
        "residuals":     np.array(fit.rx2("residuals")),
        "fitted_values": np.array(fit.rx2("fitted.values")),
        "rweights":      np.array(fit.rx2("rweights")),
        "r_squared":     fit.rx2("r.squared")[0],
    }
```

5.3. Validation Results

The following results demonstrate that the Python implementations match the R originals. All comparisons use the `flour` and `alcohol` datasets from `RobStatTM`.

5.4. Test 1: `locScaleM`, Bisquare and Huber Families

Quantity	R value	Python value	$\lvert\Delta\rvert$	Status
Bisquare, eff = 0.95, alcohol data				
$\hat{\mu}$	3.6970	3.6970	$< 10^{-4}$	PASS
$\mathrm{std}(\hat{\mu})$	0.2521	0.2521	$< 10^{-3}$	PASS
disper	0.8633	0.8633	$< 10^{-4}$	PASS
Huber, eff = 0.95, alcohol data				
$\hat{\mu}$	3.5828	3.5828	$< 10^{-4}$	PASS
$\mathrm{std}(\hat{\mu})$	0.2482	0.2482	$< 10^{-3}$	PASS
disper	0.8633	0.8633	$< 10^{-4}$	PASS

5.5. Test 2: `scaleM`, Bisquare and Huber Families

Family	R value	Python value	$\lvert\Delta\rvert$	Status
Bisquare (flour data)	matches	matches	$< 10^{-4}$	PASS
Huber (flour data)	matches	matches	$< 10^{-4}$	PASS

5.6. Additional Wrapped Functions (via rpy2)

The comparison notebook also validates all five core functions by calling R via `rpy2` and checking outputs:

Function	Dataset	Validation	Status
`lmrobdetMM`	mineral	Coefficients, scale, residuals, rweights	PASS
`covRobMM`	wine	Center vector, covariance matrix, distances	PASS
`covRobRocke`	wine	Center vector, distances	PASS
`pcaRobS`	bus	Proportion of variance, eigenvectors	PASS

Selected numerical outputs from the rpy2 wrapper:

Function	Output Field	Value (via robstatpy)
lmrobdetMM (mineral, mopt, eff = 0.95)		
	Intercept	15.2012
	Slope (copper)	0.0126
	Scale	9.9966
	Robust R^2	0.0082
covRobMM (wine, 13 variables)		
	Center[1..5]	13.74, 2.01, 2.45, 17.04, 106.34
	Max robust distance	6.09
covRobRocke (wine, 13 variables)		
	Center[1..5]	13.77, 2.03, 2.44, 17.16, 107.52
	Max robust distance	5.79
pcaRobS (bus, 3 components, desprop = 0.99)		
	propSPC[1]	> 0.80
	propex	> 0.85

All tests pass: 15/15 checks in the subprocess notebook and 14/14 in the rpy2 notebook. Both notebooks are available in the project repository: `robstatpy_comparison.ipynb` (subprocess) and `robstatpy_comparison_rpy2.ipynb` (rpy2).

6. Alternative Implementation Strategies

Two additional approaches were implemented as cross-checks and to demonstrate different trade-offs in R–Python interoperability. Both are described briefly here; full code is available in the project notebooks.

6.1. Strategy B: Subprocess + JSON Wrappers

As a comparison baseline, the project also implements wrappers using `subprocess` calls to `Rscript` with `jsonlite`-serialised results. This approach:

- Requires no compiled `rpy2` dependency (useful on systems where building `rpy2` is difficult).
- Incurs higher per-call overhead due to R process startup.

- Serves as a cross-check: if both wrappers produce identical outputs, we have high confidence in the conversion layer.

6.2. Strategy C: Direct Python Re-implementation

For `locScaleM` and `scaleM`, the project provides pure-Python re-implementations to serve as the GSoC test tasks and to establish a methodology for future direct porting.

7. Future Work: PCRA Package Conversion

The **PCRA** (Portfolio Construction and Risk Analysis) R package, also authored by Prof. Martin, depends on `RobStatTM` for robust covariance inputs to portfolio optimisation and for robust single-factor model fitting. A Python equivalent of PCRA is a natural downstream application of `robstatpy`.

7.1. Dependency Relationships

PCRA orchestrates 12+ R packages. The critical dependency chain is:

$$\begin{tikzcd}
\boxed{\text{PCRA}} \arrow[r, "\text{covRobMM}"] \arrow[d] & \boxed{\text{RobStatTM}} \arrow[r, "\text{pyinit}"] & \boxed{\text{Fortran core}} \\
\boxed{\text{PortfolioAnalytics}} \arrow[r] & \boxed{\text{PerformanceAnalytics}} \arrow[r] & \boxed{\text{xts/zoo}}
\end{tikzcd}$$

For the Python side, the mapping is:

R Package	Python Equivalent	Notes
RobStatTM	`robstatpy`	This project
PortfolioAnalytics	`cvxpy`	Convex optimisation
PerformanceAnalytics	`empyrical`	Risk/performance measures
xts / zoo	`pandas`	Time series indexing
ggplot2	`matplotlib`	Plotting

7.2. Preliminary Results

A prototype Python reproduction of PCRA Chapter 2 has already been completed (`pcra/python/ch2_foundatio` demonstrating:

- Two-asset efficient frontier (Merton closed-form).
- Global minimum variance portfolio computation.
- Drawdown analysis and risk measures (VaR, ES).
- Robust location estimation via bisquare ψ-function.

Code example: efficient frontier computation (from the prototype):

Listing 2: PCRA Chapter 2: Merton efficient frontier in Python

```python
def math_efront_risky_mu_cov(mu_ret, vol_ret, corr_ret,
                             npoints=100):
    """Efficient frontier via closed-form Merton formula.
    Reproduces R's mathEfrontRiskyMuCov() from PCRA."""
    V = np.diag(vol_ret) @ corr_ret @ np.diag(vol_ret)
    Vinv = np.linalg.inv(V)
    one = np.ones(len(mu_ret))
```

```
    a = mu_ret @ Vinv @ one
    b = mu_ret @ Vinv @ mu_ret
    c = one @ Vinv @ one
    d = b * c - a ** 2
    mu_gmv = a / c
    sigma_gmv = 1 / np.sqrt(c)
    mu_range = np.linspace(mu_gmv, max(mu_ret)*1.2, npoints)
    sigma_range = np.sqrt(
        (c*mu_range**2 - 2*a*mu_range + b) / d)
    return mu_range, sigma_range, mu_gmv, sigma_gmv
```

Selected PCRA Chapter 2 results (Python vs. R):

Quantity	R	Python	Match
GMV σ_P (2-asset, $\rho = 0$)	0.1200	0.1200	✓
GMV μ_P (2-asset, $\rho = 0$)	0.0610	0.0610	✓
3-asset GMV return	0.0540	0.0540	✓
3-asset GMV volatility	0.0707	0.0707	✓
VaR (5%, SPY monthly)	matches	matches	✓
ES (5%, SPY monthly)	matches	matches	✓
Max drawdown depth	matches	matches	✓

The complete comparison output (11 figures, 8 tables) is available in the project repository: pcra/output/Ch2_Comparison.pdf (R vs. Python side-by-side). These results match the R originals and validate the feasibility of converting PCRA's core mathematical layer to Python. Full PCRA conversion is scoped as a stretch goal; the foundation laid by `robstatpy` makes it achievable in a follow-up contribution.

The `rpy2` wrapper methodology scales directly to PCRA and other R packages in the robust statistics ecosystem: once conversion patterns are established for `RobStatTM`, extending them to PCRA is incremental. Full PCRA conversion is scoped as a stretch goal, contingent on completing the core `robstatpy` deliverables.

8. Evaluation Plan: Comparative Benchmarks

A central deliverable is a rigorous comparison of the three implementation strategies across three dimensions:

8.1. Metrics

1. **Accuracy.** For each function, compute $\max_i |y_i^{\mathrm{Py}} - y_i^{\mathrm{R}}|$ across all return fields, for both the rpy2 wrapper (expected: 0) and the pure-Python port (expected: $< 10^{-6}$).
2. **Speed.** Wall-clock time per call, measured with `timeit` over 100 repetitions on synthetic data of sizes $n \in \{100, 1000, 10000\}$ and $p \in \{5, 20, 50\}$. Breakdown: R computation time vs. conversion overhead.
3. **Ergonomics and integration cost.** Lines of glue code per function, dependency footprint, and ease of installation (does the approach require R to be installed?).

8.2. Benchmark Protocol

- **Synthetic data generation:** Gaussian base with $\varepsilon = 5\%$ contamination at 10σ to exercise breakdown behaviour.
- **Functions benchmarked:** `locScaleM`, `scaleM`, `lmrobdetMM`, `covRobMM`, `covRobRocke`, `pcaRobS`.
- **Comparison grid:** rpy2 wrapper vs. subprocess wrapper vs. pure-Python port (where available) vs. nearest `scikit-learn`/`statsmodels` equivalent (where one exists).
- **Reporting:** Results will be presented in a Jupyter notebook with timing tables, accuracy heat maps, and commentary.

9. Implementation Plan and Timeline

The project follows a **22-week large-project timeline** (350 hours), structured around the GSoC 2026 calendar.

9.1. Community Bonding Period (May 1 – May 24)

- Finalise package scaffolding: `pyproject.toml`, `pytest` configuration, CI/CD with GitHub Actions (R + Python matrix).
- Set up automated testing: R installation in CI, `rpy2` build, dataset fixtures.
- Draft the `robstatpy` API surface with type stubs and docstring templates.
- Agree on code review cadence and milestone criteria with mentors.

9.2. Phase 1: Core Univariate Wrappers (Weeks 1–3, May 25 – Jun 14)

Week	Deliverables
1	`locScaleM` wrapper + pure-Python port + pytest suite (bisquare, huber, mopt families; eff $\in \{0.85, 0.90, 0.95\}$). Numerical agreement verified to $< 10^{-8}$.
2	`scaleM` / `mscale` wrapper + pure-Python port. Subprocess wrapper as cross-check. Document relationship between `scaleM` and `locScaleM`.
3	`lmrobdet.control` wrapper exposing all tuning parameters. Comparison notebook: rpy2 vs. subprocess vs. pure-Python for univariate functions.

9.3. Phase 2: Robust Regression (Weeks 4–7, Jun 15 – Jul 12)

Week	Deliverables
4	`lmrobdetMM` wrapper: formula interface, coefficient extraction, residuals, robustness weights, scale.
5	`lmrobdetMM` diagnostics: robust R^2, fitted values, summary table mirroring R's `summary()` output. `lmrobdetMM.RFPE` wrapper.
6	`step.lmrobdetMM` wrapper for stepwise robust regression. Reproduce mineral dataset examples (Figs 5.1–5.7).
7	**Midterm milestone:** complete regression module with full pytest coverage. Prepare midterm evaluation report.

Midterm Evaluation (Jul 6–10): Deliverables: univariate wrappers, regression wrappers, comparison notebook, $\geq 90\%$ test coverage on delivered modules.

9.4. Phase 3: Robust Covariance and PCA (Weeks 8–11, Jul 13 – Aug 9)

Week	Deliverables
8	`covRobMM` wrapper: center, covariance, correlation, Mahalanobis distances, weights.
9	`covRobRocke` wrapper. Reproduce wine dataset examples (Fig 6.3).
10	`pcaRobS` wrapper: eigenvectors, proportion of variance, scores. Reproduce bus dataset examples (Fig 6.10).
11	`prcompRob` wrapper (convenience wrapper over `pcaRobS`). Distance–distance plot utility function. Scree plot utility function.

9.5. Phase 4: Benchmarks and Comparative Study (Weeks 12–14, Aug 10 – Aug 30)

Week	Deliverables
12	Speed benchmarks: rpy2 vs. subprocess vs. pure-Python vs. scikit-learn/statsmodels baselines. Synthetic data generator for contaminated Gaussian.
13	Accuracy benchmarks: max absolute error across all functions, all parameter combinations. Tabulate conversion overhead breakdown.
14	Write-up: comparative analysis Jupyter notebook with tables, plots, and recommendations for when each approach is appropriate.

9.6. Phase 5: Documentation, Packaging, and Polish (Weeks 15–17, Aug 31 – Sep 20)

Week	Deliverables
15	Complete API documentation (Sphinx + ReadTheDocs). Every function has a docstring modelled on the R man page.
16	Publish `robstatpy` on PyPI (sdist + wheel). Installation guide covering R dependency setup on Linux/macOS/Windows.
17	Tutorial Jupyter notebooks reproducing key textbook chapters (Chapters 2, 5, 6 of Maronna et al.).

9.7. Phase 6: PCRA Extension and Final Deliverables (Weeks 18–22, Sep 21 – Nov 2)

Week	Deliverables
18–19	**PCRA exploration:** Map PCRA's dependency on `RobStatTM` (calls to `locScaleM`, `lmrobdetMM` in portfolio risk contexts). Wrap or port PCRA's core math functions (efficient frontier, global minimum variance).
20	PCRA comparison notebook: R vs. Python outputs for Chapter 2 demo (2-asset frontier, drawdown analysis, risk measures).
21	Integration testing: end-to-end portfolio analytics workflow using `robstatpy` for robust covariance input.
22	**Final submission:** Final code review, README polish, contributor guide, final evaluation report.

10. Deliverables

10.1. Required Deliverables

1. **robstatpy Python package** (pip-installable) wrapping the following RobStatTM functions via rpy2:
   - locScaleM, scaleM (univariate)
   - lmrobdetMM, lmrobdet.control, lmrobdetMM.RFPE, step.lmrobdetMM (regression)
   - covRobMM, covRobRocke (covariance)
   - pcaRobS, prcompRob (PCA)
2. **Pure-Python re-implementations** of locScaleM and scaleM with verified numerical equivalence, establishing the methodology for progressively porting the full library to native Python.
3. **Pytest test suite** achieving $\geq 90\%$ coverage on all wrapper functions, comparing Python outputs against R ground truth.
4. **Comparative benchmark notebook** evaluating speed, accuracy, and ergonomics across rpy2 wrappers, subprocess wrappers, and pure-Python ports.
5. **Sphinx documentation** published on ReadTheDocs, with docstrings modelled on the R man pages.
6. **Tutorial Jupyter notebooks** reproducing textbook examples from Chapters 2, 5, and 6 of Maronna et al. (2019).

10.2. Stretch Deliverables

1. **PCRA Python module** wrapping or porting core portfolio analytics functions (mathEfrontRiskyMuCov, mathGmv, drawdown analysis, risk measures).
2. **Additional RobStatTM wrappers:** lmrobM, BYlogreg (Bianco–Yohai logistic regression), rob.linear.test.
3. **Complete native Python port** of the remaining RobStatTM functions (lmrobdetMM, covRobMM, covRobRocke, pcaRobS), validated against the rpy2 wrappers, with optional numba JIT acceleration for performance-critical inner loops.

11. Distribution and Packaging

robstatpy will be distributed through two complementary channels to maximise accessibility:

11.1. GitHub Installation (Development and Bleeding-Edge)

The package will be hosted on GitHub under an open-source license (MIT or Apache 2.0, to be confirmed with mentors). Users can install directly from the repository:

Listing 3: GitHub installation

```
pip install git+https://github.com/Aakarsh751/robstatpy.git
```

This channel provides access to the latest development version and makes it straightforward for contributors to fork, modify, and submit pull requests. The repository will include:

- A `pyproject.toml` with full metadata and dependency specifications (`rpy2 >= 3.5`, `numpy`, `pandas`, `scipy`, `matplotlib`).
- GitHub Actions CI running the full `pytest` suite on Linux, macOS, and Windows against multiple R versions (4.3, 4.4, 4.5).
- A `CONTRIBUTING.md` with guidelines for adding new wrappers or submitting pure-Python ports.
- Tagged releases corresponding to each PyPI publication.

11.2. PyPI Publication (Stable Releases)

Stable, versioned releases will be published on the Python Package Index (PyPI), enabling standard installation:

Listing 4: PyPI installation

```
pip install robstatpy
```

The packaging workflow uses:

- `build` to produce source distributions (`sdist`) and pure-Python wheels.
- `twine` for secure upload to PyPI.
- Semantic versioning (e.g., `0.1.0` for the initial GSoC release, `0.2.0` after PCRA extension).
- A GitHub Actions release workflow that automatically publishes to PyPI when a new git tag is pushed.

R dependency handling: Since `robstatpy` requires a working R installation with `RobStatTM`, the package will: (1) detect `R_HOME` automatically at import time, (2) provide a clear `ImportError` message with installation instructions if R is not found, and (3) include a `robstatpy.check_setup()` utility that validates the R environment and prints version information. The installation guide will cover R setup on all three major platforms (Linux via `apt/conda`, macOS via Homebrew, Windows via the CRAN installer).

12. Risks and Mitigation Strategies

Risk	Impact	Mitigation
`rpy2` build failures on Windows/macOS	Delays CI and user installation	Provide pre-built wheels where possible; subprocess wrapper as fallback; Docker image for reproducible builds
R version incompatibility	`RobStatTM` may behave differently across R versions	Pin minimum R version (4.3+); test in CI against 4.3, 4.4, 4.5
Numerical discrepancies in pure-Python ports	Undermines credibility of direct-porting approach	Tolerance tiers: 10^{-12} for wrappers, 10^{-6} for ports; investigate any discrepancy $> 10^{-4}$ before merging
Scope creep from PCRA extension	Core `robstatpy` quality suffers	PCRA is explicitly a stretch goal; only begin after Phase 4 benchmarks are complete and midterm evaluation is passed
Mentor availability during summer	Reduced feedback cadence	Three mentors provide redundancy; asynchronous communication via GitHub issues; weekly status updates regardless

13. References

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[3] V. J. Yohai. "High breakdown-point and high efficiency robust estimates for regression." *The Annals of Statistics*, 15(2):642–656, 1987.

[4] D. L. Rocke. "Robustness properties of S-estimators of multivariate location and shape in high dimension." *The Annals of Statistics*, 24(3):1327–1345, 1996.

[5] L. Gautier et al. `rpy2`: R in Python. https://rpy2.github.io/, 2024.

[6] R. D. Martin. `PCRA`: Portfolio Construction and Risk Analysis. CRAN, 2023.